

DAVULURI YASHWANT RAO

Full Stack Developer

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PROFESSIONAL SUMMARY

Full Stack Developer with 2+ years of hands-on experience designing, developing, and deploying production-grade web applications. Proficient in React, Next.js, Node.js, Python, FastAPI, and React Native, with experience across frontend architecture, backend development, REST APIs, authentication and authorization, databases, and cloud deployment. Strong track record of building end-to-end applications, improving performance and delivering scalable solutions in production environments.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: Next.js, React.js, React Native, SvelteKit, Svelte, Redux toolkit

Backend: Node.js, Express, FastAPI, Django, Python, Flask

Databases & Storage: PostgreSQL, MongoDB, Firebase, AWS S3, SQLite, DuckDB, MySQL

Infrastructure & DevOps: Docker, Linux, AWS, CI/CD, Git, Postman

APIs & Architecture: REST APIs, GraphQL, JWT Auth, OAuth 2.0, RBAC, Socket.io, Agentic AI, MCP Tools

QA & Testing: Playwright

PROFESSIONAL EXPERIENCE

Full Stack Developer I | Aganitha Cognitive Solutions

Feb 2025 – Present

- **Optimized Python computational pipelines reducing execution time from 20 hours to 4 hours per unit (80% improvement)**

Implemented advanced algorithm optimization and efficient resource allocation for data-intensive processing tasks

- **Built Next.js research data platform from scratch enabling users to save, compare, and validate results against public data sources**

- Architected PostgreSQL backend with Docker containerization and AWS deployment
- Optimized critical performance bottleneck: reduced data retrieval from 40 seconds using request chunking and IndexedDB client-side caching
- Implemented on-demand and 24-hour automated data refresh preventing stale data while minimizing API calls
- Deployed JWT authentication and role-based access control (RBAC) enabling admin data management and user permissions

- **Engineered SvelteKit scientific tool platform over 1.4+ years, evolving from beta-stage to production application**

- Implemented CLI tooling, admin interfaces for formula/algorithm management with PostgreSQL and AWS deployment
- Deployed user and admin role-based access control with JWT authentication
- Continuously delivered feature enhancements, bug fixes, and maintenance updates across full application lifecycle

MERN Full Stack Intern | MyParticipants

Aug 2024 – Nov 2024

- **Developed full-stack features using React, TypeScript, and Node.js with end-to-end ownership from design through production deployment**

- **Implemented JWT and Google OAuth authentication with token-based collaborator access control**
 - Enabled secure multi-user collaboration and role-based permissions within MERN application
- **Built comprehensive file management system supporting PDF handling, image processing, report generation, and optimized file uploads**

SELECTED PROJECTS

Xchange – Student Marketplace with Real-Time Communication

React Native | Node.js | Express | Socket.io

- Mobile marketplace enabling peer-to-peer item exchange and commerce within college and hostel communities
- Implemented real-time chat using Socket.io for instant user communication and transaction coordination

VR/AR Real Estate Platform

MERN Stack | Firebase | 3D Visualization | AR/VR

- Real-estate platform with immersive 3D property tours and AR furniture visualization for pre-visit property assessment
- Enabled users to virtually place furniture inside properties to determine fit and spatial arrangement
- **Research published in IEEE paper**

Xchange Web – College Community Platform

Next.js | GraphQL | MongoDB

- Purpose-built community platform for college students with community creation, public/private chatrooms, and news sharing
- GraphQL-based data architecture for optimized data fetching and real-time collaboration features

Agentic AI Automation System

Node.js | Next.js | AI Agents | Workplace Automation

- AI agent system automating multi-step workplace workflows including email automation, meeting scheduling, and timesheet management
- Practical Agentic AI implementation focused on executable task automation beyond chatbot interfaces

EDUCATION

B.Tech, Computer Science & Engineering

Vel Tech University | 2021 – 2025 | **CGPA: 8.7**

SOCIALS

LinkedIn: www.linkedin.com/in/yashwant-rao-54a506250

GitHub: <https://github.com/Y-A-S-H-W-A-N-T>

Portfolio: <https://yashwant-portfolio.vercel.app/>